

Kush Singhal

Curriculum Vitae

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Education

- 2022–2027 **Ph.D.**, *Harvard University*, Cambridge, MA, (expected)
Advisor: Mark Kisin
- 2018–2022 **B.Sc.**, *University of Hong Kong*, Hong Kong

Publications/Preprints

1. Kush Singhal. *Coleman Isomorphisms in Syntomic Cohomology and THH*. 2026. arXiv: 2609.05252 [math.AG]. URL: <https://arxiv.org/abs/2609.05252>

Undergraduate Papers:

2. Kush Singhal. *The Completed L -function attached to the Weight 2 Polar Harmonic Maass Form $H_{N,z}^*(\tau)$* . 2022. arXiv: 2201.03146 [math.NT]. URL: <https://arxiv.org/abs/2201.03146>
3. Kush Singhal. “Near-miss identities and spinor genus classification of ternary quadratic forms with congruence conditions”. In: *International Journal of Number Theory* 20.04 (2024), pp. 951–993

Seminar Organising

- Summer 2026 **Analytic Syntomification Learning Seminar**, Harvard
- Fall 2024, **Harvard Number Theorists’ Seminar**, Harvard
- Fall 2025
- Fall 2023, **Trivial Notions Seminar**, Harvard
- Spring 2024

Teaching Experience

- Fall 2026 **Coach**, Harvard
Course: Math 21a, Introduction to Multivariable calculus
- Spring 2026 **Course Assistant**, Harvard
Course: Math 223b, Galois Cohomology and Class Field Theory
- Spring 2026 **Mentor**, *Directed Reading Program*, Harvard
- Fall 2025 **Teaching Fellow**, Harvard
Course: Math 1a, Introduction to Calculus
- Fall 2025 **Mentor**, *Directed Reading Program*, Harvard
- Spring 2025 **Mentor**, *Directed Reading Program*, Harvard
- Fall 2024 **Teaching Fellow**, Harvard
Course: Math 1a, Introduction to Calculus

- Fall 2024 **Mentor**, *Directed Reading Program*, Harvard
 Spring 2024 **Mentor**, *Directed Reading Program*, Harvard
 Fall 2023 **Teaching Fellow**, Harvard
Course: Math 1a, Introduction to Calculus
 Fall 2023 **Mentor**, *Directed Reading Program*, Harvard
 Summer 2023 **Lecturer**, *Summer Tutorial*, Harvard
Topic: Arithmetic Theory of Quadratic Forms
 Spring 2023 **Mentor**, *Directed Reading Program*, Harvard
 Fall 2023 **Mentor**, *Directed Reading Program*, Harvard
 Spring 2022 **Student Teaching Assistant**, HKU
Course: Math 2102, Linear algebra II
 Spring 2021 **Student Teaching Assistant**, HKU
Course: Math 2012, Fundamental Concepts of Mathematics
 Fall 2020 **Student Teaching Assistant**, HKU
Course: Math 2241, Introduction to Mathematical Analysis

Talks

Research Talks

- Oct 2026 **Coleman Isomorphisms in THH**, (*upcoming*)
 JHU Topology Seminar, Johns Hopkins University
 Apr 2026 **Remarks on Iwasawa Theory via $K(1)$ -Local K Theory**
 Harvard Number Theorists' Seminar, Harvard
 Oct 2025 **Iwasawa Theory and $K(1)$ -Local Homotopy Theory**
 Zygotop Seminar, Harvard
 Sep 2025 **Non-Triviality of Euler Systems via Homotopy Theory**
 Harvard Number Theorists' Seminar, Harvard
 Sep 2024 **Explicit Reciprocity Laws via Prismatic Cohomology**
 Harvard Number Theorists' Seminar, Harvard
 Jul 2021 **Computing Spinor Genera of Ternary Quadratic Forms with Congruence Conditions**
 Number Theory Days 2021, HKU

Expository

- Aug 2026 **The Analytic Prismaticisation**
 Analytic Syntomification Learning Seminar, Harvard
 Apr 2026 **Higher Cyclotomic Extensions**
 Thursday Seminar, Harvard
 Mar 2026 **TC, TP, and All That**
 Babytop, Harvard
 Feb 2026 **K -Theoretic Artin Maps**
 The p -Adic Seminar, Harvard

- Dec 2025 **Abstract Traces, and Hochschild Homology of Group Algebras**
Babytop, Harvard
- Oct 2025 **“Origamists Don’t Want You to Know This One Weird Trick”**
Trivial Notions, Harvard
- Apr 2025 **The Nygaard Stack for Semi-Perfectoids**
Learning Seminar on Prismatic F -Gauges, Harvard
- Apr 2025 **Counting Rational Points on Curves**
Trivial Notions, Harvard
- Apr 2025 **The $K(1)$ -local sphere**
Babytop, Harvard
- Sep 2024 **Can you fold the cube root of 2?**
Trivial Notions, Harvard
- Mar 2024 **Hodge–Tate Comparison in Mixed Characteristic**
Learning Seminar on Prismatic Cohomology, Harvard
- Mar 2024 **A Climb Up the Tower**
Trivial Notions Seminar, Harvard
- Sep 2023 **Can you (flat-)fold it?**
Trivial Notions Seminar, Harvard
- Mar 2023 **Review of Reductive Algebraic Groups**
Seminar on Topics in Arithmetic, Geometry, Etc., MIT
- Oct 2022 **How many integer solutions are there to $2x^2 + 2y^2 + 2z^2 + x = 190$?**
Trivial Notions Seminar, Harvard
- Nov 2022 **An Example of Descent**
Seminar on Topics in Arithmetic, Geometry, Etc., MIT

Selected Honours and Awards

- 2020–21 **B.Sc. Class of 1971 Prize**, *Faculty of Science*, HKU
- 2020–21 **Hong Kong University Alumni Award**, *Faculty of Science*, HKU
- 2020–21 **Dean’s Honours List**, *Faculty of Science*, HKU
- 2019–20 **Dean’s Honours List**, *Faculty of Science*, HKU
- 2018–19 **Dean’s Honours List**, *Faculty of Science*, HKU